

BIRD SOUND CLASSIFICATION

using MFCC & SVM

Team Members: BEN FLIS ZIYA
THANHA NOORUDHEEN TT
ALEENA V J



PROBLEM STATEMENT & MOTIVATION



THE NEED

Monitor biodiversity at scale using audio - non-invasive, continuous, and deployable anywhere.



THE CHALLENGE

Distinguish similar-sounding species like Cuckoo vs. Kingfisher from messy background noise.

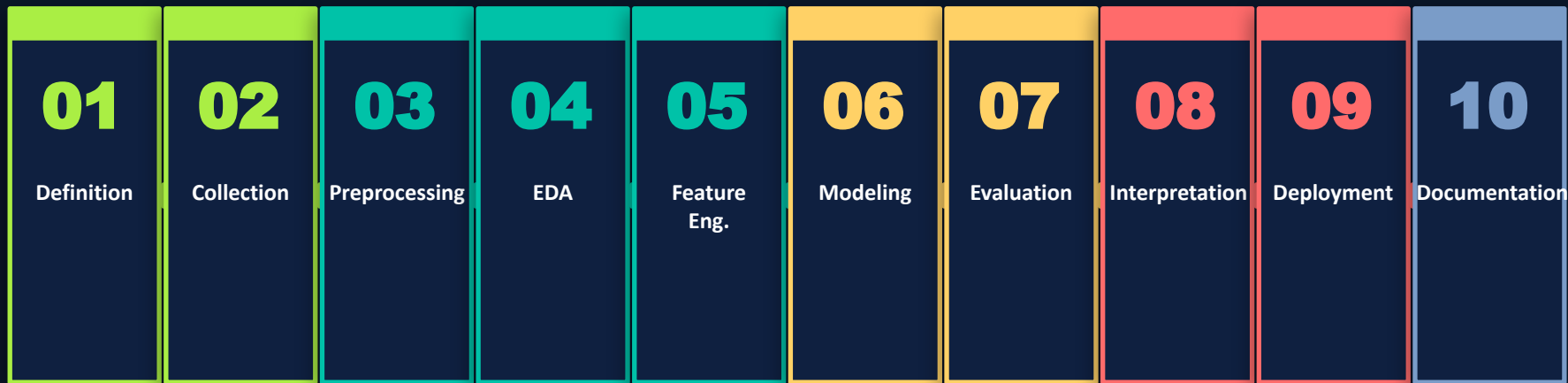


OUR APPROACH

Classical ML (SVM) over Deep Learning - resource-efficient, interpretable, runs on any device.

DATA SCIENCE

LIFE CYCLE



DATASET OVERVIEW

347

Total
Samples

5

Bird
Classes

140

Features
per File

CLASS DISTRIBUTION



Source: Xeno-Canto / Kaggle

FEATURE ENGINEERING

MFCC

20 coeff $\times$ 4 stats

Simulates human hearing, captures timbre & pitch.

80

Spectral Centroid

1 feat $\times$ 4 stats

Brightness : sharp chirp vs. dull call.

4

Spectral Rolloff

1 feat $\times$ 4 stats

Separates noise from tonal bird calls.

4

Zero Crossing Rate

1 feat $\times$ 4 stats

Detects percussive / noisy bursts.

4

Chroma Features

12 bins $\times$ 4 stats

Harmonic & melodic pitch content of call.

48

MODEL COMPARISON

★ WINNER

SVM

RBF · C=10 · γ =scale

81.4%

accuracy

F1: 0.80

Best performer, used for deployment

RF

200 trees · seed=42

~78%

accuracy

F1: 0.77

Provides feature importance insights

KNN

k=5 · Euclidean

~72%

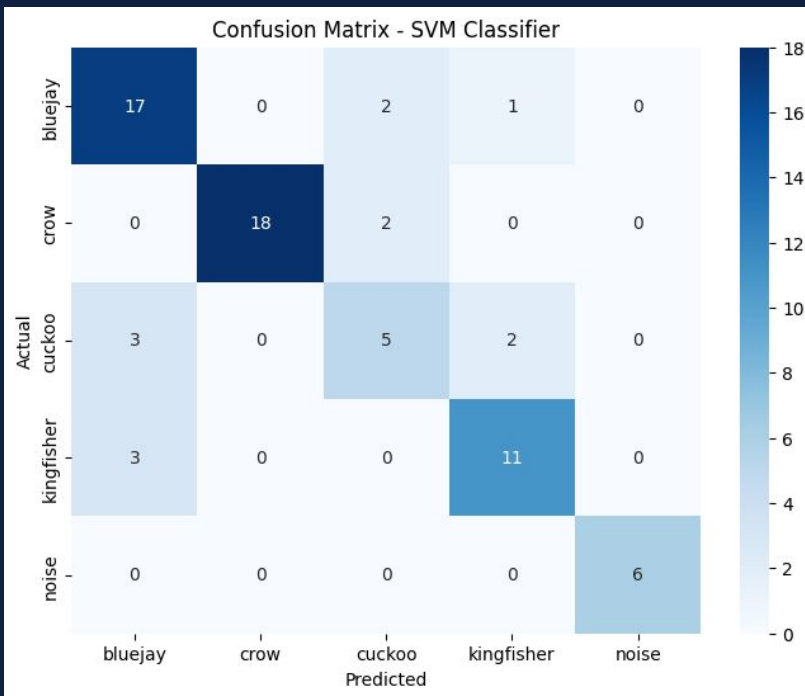
accuracy

F1: 0.70

Simple, fast baseline classifier

MODEL EVALUATION

CONFUSION MATRIX



KEY INSIGHTS

Noise

Highest accuracy : unique spectral signature

Crow

Strong recall : low-frequency raspy call

Bluejay

Slight confusion with Kingfisher

Cuckoo

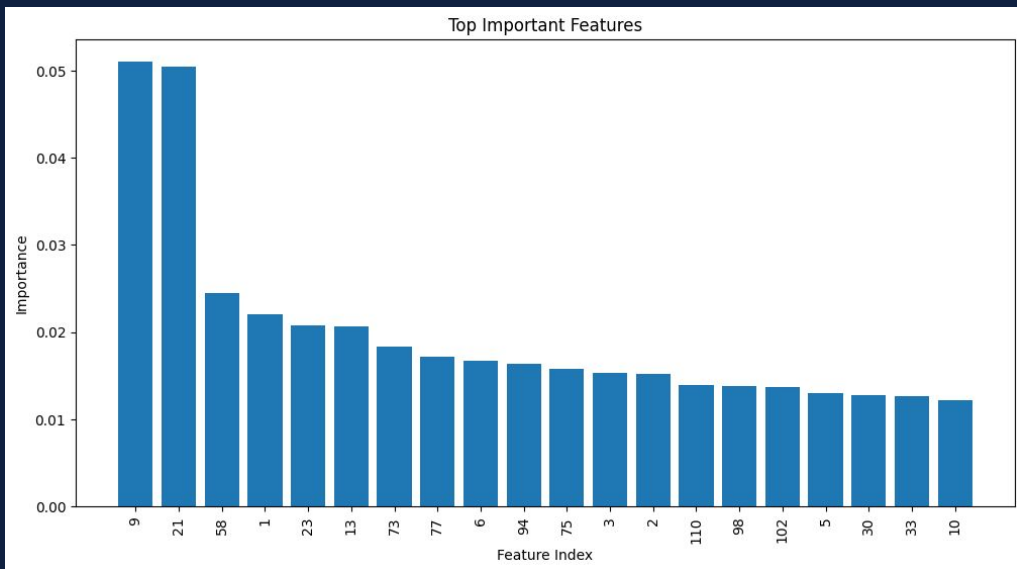
Trickiest : varying pitch causes mix-ups

Kingfisher

Good recall : sharp whistle stands out

FEATURE IMPORTANCE

EXPLAINABILITY — RANDOM FOREST



FEATURE INDEX MAP

0 – 79

MFCC stats

80 – 83

Spectral Centroid

84 – 87

Spectral Rolloff

88 – 91

Zero Crossing Rate

92 – 139

Chroma Features

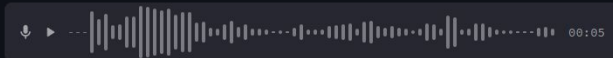
STREAMLIT DEPLOYMENT

Bird Sound Classification

Record or upload bird audio to identify bird species.

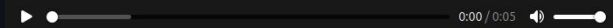
Record Bird Sound

Record Audio



Or Upload Audio File

 Upload 200MB per file • WAV, MP3



Predicted Bird: cuckoo

Supported Classes:

crow, cuckoo, kingfisher, bluejay, noise



Upload .wav or .mp3 directly



Record audio via microphone



Instant SVM prediction



MFCC visual + confidence score



Live URL:
<https://bird-sound-classification-mqczgejumcx4e5u7sxb5jt.streamlit.app/>

model/bird_svm_model.pkl • model/scaler.pkl • app.py

CONCLUSION

& FUTURE WORK

✓ MFCC + SVM achieved 81.4% accuracy, proving classical ML is highly effective for bio-acoustic classification with limited data.

⚠ LIMITATIONS

- Small dataset, Cuckoo underrepresented.
- No real-time field recording pipeline yet.
- Limited background noise diversity.

🚀 FUTURE WORK

- CNN on mel-spectrograms for higher accuracy.
- Real-time Raspberry Pi mic pipeline.
- Federated community bird monitoring.
- Expand to 20+ species from Xeno-Canto.